

Aryaroop Majumder

AI Engineer

Bengaluru, India | +91 97848 07414 | aryaroop04@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

AI Engineer and GenAI/LLM-focused backend engineer with 3+ years of consulting/industry experience and 4+ years of Python across professional, academic, and open-source AI/data projects. Experienced designing AI-powered applications and GenAI architectures that integrate LLMs, RAG, embeddings, vector databases, document ingestion, APIs, AI pipelines, and agent-based components for enterprise use cases. Hands-on with Python, FastAPI, Flask, REST APIs, SQL, PySpark, Docker, CI/CD, GCP, AWS, Azure OpenAI, Hugging Face, Ollama, quantized models, prompt engineering, MCP, LLM evaluation, and production-oriented delivery.

CORE SKILLS

Backend, APIs and Microservices: Python, FastAPI, Flask API, REST APIs, asyncio concepts, SQL, Pandas, PySpark, API design, backend data processing, microservice-oriented patterns

GenAI, LLMs and NLP: GPT-style LLMs, LLaMA, Mistral/Phi familiarity, Hugging Face Transformers, Ollama, quantized models, prompt engineering, system prompts, few-shot prompting, evaluation prompts, summarization, Q&A, topic modelling, semantic search, text classification, entity extraction

RAG, Knowledge Engineering and Evaluation: RAG architectures, RAG knowledge base creation, document embeddings, chunking, metadata tagging, hybrid search, BM25, reranking, grounding validation, hallucination detection concepts, LLM evaluation frameworks, retrieval quality evaluation, R@K/MRR/context relevancy concepts

Vector Stores and Data Sources: Pinecone, FAISS, Qdrant, vector databases, PostgreSQL foundations, OCR, PDFs, Word, PPT, HTML, images, structured/unstructured ingestion, ETL connectors, data modeling for AI workloads

Azure AI, Cloud and MLOps: Azure OpenAI, Azure AI Search/Cognitive Search concepts, Azure Blob Storage, Azure VM, Azure ML/Data Lake familiarity, AWS Lambda, AWS Glue, GCP, Google Cloud Build, GitHub Actions, Docker, Airflow, CI/CD, MLOps pipeline exposure, logging/latency/cost monitoring concepts

UI, Enterprise Delivery and Responsible AI: React.js, Angular, Streamlit, product demos, PoCs, architecture reviews, best-practice documentation, stakeholder communication, security/compliance awareness, privacy, bias/fairness/transparency awareness

PROFESSIONAL EXPERIENCE

Data Science Specialist | Brainlabs | Bengaluru, India | April 2025 - Present

- Designed and delivered **SearchIQ** and an AI SEO product suite combining GenAI utilities, a **RAG-based chatbot**, enterprise document/content grounding, SEO analytics, reusable data-processing workflows, and **Google Cloud Build CI/CD** support, reducing SEO analysis turnaround from months to weeks and cutting analysis time by **85%**.
- Implemented **SearchIQ** RAG and knowledge-retrieval components using document embeddings, chunking, metadata tagging, hybrid retrieval patterns, **Pinecone/FAISS** vector stores, **Hugging Face** open-source models, **Ollama**-hosted models, and **quantized model variants**, improving keyword analysis accuracy to **95%**.
- Used **LLM evaluation frameworks** and retrieval-quality checks across **SearchIQ** workflows to assess groundedness, context relevancy, hallucination risk, and answer quality, while reducing platform cost from **\$500+** to **\$100-\$150** for **200+ users** through model, retrieval, and infrastructure optimization.
- Built GenAI product-tagging automation for a **Fortune 500** global consumer products client by combining image signals, onsite content scraping, prompt engineering, classification logic, and validation workflows to reduce manual categorization effort and create a scalable content intelligence workflow for client teams.
- Developed a client-agnostic topical classification module for an enterprise SEO analytics initiative, categorizing search phrases across heterogeneous client sources after rapid solution evaluation, technical grooming, architecture option analysis, and stakeholder review, reducing classification cost by **60%** and turnaround time by **30%**.
- Delivered AI-enabled analytics utilities including an embedding analyzer with topic clustering, SERP feature analyzer, keyword volume analyzer, cross-linkage matching tool, and Screaming Frog OSS alternative to accelerate semantic search, SEO diagnostics, and content opportunity discovery.
- Prepared business one-pagers, executive decks, demo flows, and **SearchIQ** narratives for senior stakeholders, strategic partner pitch opportunities, **Fortune 500** consumer brand pursuits, marketing mix modeling PoCs, GenAI solutioning conversations, and best-practice guidance discussions.
- Led **Docker**, **GitHub Actions**, **Airflow**, **LangGraph**, and AI knowledge-sharing sessions, supported an agentic AI orchestration use case using **LangGraph** instead of a standard ADK-style workflow, used **Airflow** for **scheduled workflow execution** around the **LangGraph** agent system, shared implementation learnings with peers, supported PyMC MMM demos, and helped convert ambiguous prototypes into repeatable workflows with clearer delivery ownership and documentation.

Associate ML Engineer | Prescience Decision Solutions | Bengaluru, India | August 2024 - March 2025

- Developed scalable **PySpark** ETL and data transformation workflows for **Amazon** using Python, Pandas, and PySpark to support backend data preparation, schema normalization, reusable transformations, and downstream AI/data workload readiness.
- Implemented advanced data quality solutions with rule-based checks, data validation workflows, exception handling, audit-friendly reporting logic, and quality controls, resolving **99.99%** of identified quality issues across supported processing workflows.
- Engineered **Amazon**-focused big data production pipelines and optimized **AWS Glue DPU** usage, improving processing performance by **35%** and reducing operational costs by **50%** while maintaining stable delivery for data-heavy workloads.

- Deployed and supported data and AI workflows across AWS Lambda, AWS Glue, **Amazon EC2**, **Amazon S3**, Azure Blob Storage, Azure VM, and **Azure OpenAI**-enabled environments, partnering with engineering teams on release support, debugging, environment readiness, and production stability.

Data Science Intern | Prescience Decision Solutions | Bengaluru, India | June 2023 - July 2024

- Built a **Flask API**-based data quality product deployed across AWS and Azure environments, then optimized backend logic, validation rules, and service performance to improve accuracy and runtime efficiency by **30%**.
- Developed an **Airflow**-based ML/data pipeline using open-source orchestration for scheduled ingestion, transformation, and model/data workflow execution, reducing recurring platform costs and increasing processing speed by 25%.
- Built a **Natural Language to SQL** chatbot with Llama 3.1, Google Flan-T5, StarCoder, Streamlit, **Flask API**, and **Docker**, enabling non-SQL users to query structured data through conversational interfaces and prototype database access workflows.

SELECTED AI AND ENGINEERING PROJECTS

- **Hybrid Contextual RAG Chatbot** | GitHub: Solely built a production-style **RAG application** with **FastAPI** endpoints for ingestion and Q&A, multi-format document ingestion for **PDFs, DOCX, XLSX, PNG/images** and other files, **PyPDF/Tesseract**-style extraction, **OCR, RAG knowledge base** creation, **Qdrant vector search**, dense retrieval, **BM25** sparse retrieval, reciprocal rank fusion, **LLM reranking**, neighboring chunk expansion, conversation-aware retrieval, grounded answers with citations, Docker deployment, **GitHub Actions CI/CD**, health checks, and local/remote vector DB support.
- **Agentic AI / Assistant Tooling** | GitHub / PyPI: Built and published **yobitsugi**, a Python-based security scanning and LLM-assisted remediation workflow supporting **Claude Code**, Codex, Cursor, Gemini CLI, Aider, **OpenAI, Anthropic**, Google, **Ollama**, configurable providers, deterministic scan/fix/validate stages, rollback logs, **GitHub Actions CI/CD**, **governance-friendly traceability**, and assistant skill installation.
- **Portfolio Website** | **React.js**: Built and maintains a **React.js**-based **portfolio website** demonstrating front-end implementation, project storytelling, visual presentation, and public proof of AI/ML, RAG, Python, Docker, and applied engineering work.
- **Dynamic Graphing System** | GitHub: Built a self-contained **Dockerized Streamlit** BI dashboard with Python service structure for interactive analytics workflows, giving users a lightweight way to explore business data and generate visual summaries without a heavy BI platform.
- **Public Portfolio and Repository Body**: Maintains a public scientific ML/data science portfolio and GitHub profile with 32 public repositories spanning RAG, Python, Docker, Streamlit dashboards, PySpark notebooks, LLM chatbots, MCP/Claude tooling, and applied ML projects.

EDUCATION

Lovely Professional University | B.Tech, Computer Science Engineering, Data Science, Machine Learning and AI | 2020 - 2024 | Phagwara, India

CERTIFICATIONS AND AWARDS

- Introduction to Model Context Protocol (Anthropic, Jun 2026)
- Certificate of Completion: Claude 101 (Anthropic, Jun 2026)
- Generative AI: Introduction and Applications (IBM, Jun 2026)
- Introduction to Artificial Intelligence (AI) (IBM, Jun 2026)
- Building Generative AI Solutions with Best Practices and Design Choices - Technical, Level 300 Advanced ML (AWS, Mar 2024)
- Foundations of Prompt Engineering (AWS, Apr 2024)
- Amazon ML Summer School 2022, Top 200 across India